

Scalability & Future Plan

Date	2 July 2026
Team ID	XXXXXX
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a small sample dataset for training	Prediction accuracy may be limited	High
2	No real-time weather or flood sensor integration	Cannot provide live flood alerts	High
3	Basic Flask application with limited scalability	Supports only a small number of users	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for a small number of users	Deploy on cloud platforms (AWS, Azure, or Google Cloud) with load balancing
2	Data Storage	Stores a small local dataset	Use cloud databases such as MySQL, PostgreSQL, or MongoDB to handle larger datasets
3	Performance	KNN model works on a limited dataset	Train the model with larger datasets and optimize prediction speed using advanced ML models
4	Security	Basic Flask application	Implement user authentication, HTTPS, and secure API access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Improve dataset size and model accuracy	1–2 Months	Better flood prediction accuracy
Phase 3	Integrate real-time weather and rainfall APIs	3–4 Months	Live flood monitoring and alerts
Phase 4	Develop Android mobile application with notifications	4–6 Months	Easy access for users and emergency alerts